



The Key Variables Needed for PFDavg Calculation

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exida
80 N. Main St.
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www.exida.com**

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Abstract

In performance based functional safety standards, safety instrumented function (SIF) designs are verified using specified metrics. A key metric for SIF designs deployed in low demand applications in the process industries is called average Probability of Failure on Demand (PFDavg). As the result of numerous studies of many field failure and proof test reports, several variables have been identified as key to a realistic PFDavg calculation. Most simplified equations including the informative section in IEC 61508, Part 6 do not include several key variables. It is shown that exclusion of these parameters can result in an optimistic PFDavg metric calculation which may result in an unsafe design.

This paper identifies the key variables that need to be included in any PFDavg calculation and provides, for the most common low demand architecture (1oo1), some simplified equations showing the impact of these key variables. A specific example of a 1oo1 architecture is twice analyzed – once with several key variables omitted and once with all key variables included. A comparison of the results of the two calculations shows that with key variables omitted the calculated PFDavg supports safety integrity level (SIL) 2 whereas with all key variables included the calculated PFDavg barely supports SIL 1!

Introduction

IEC 61511, the functional safety standard for the process industries, is “performance” based. Rather than having specific designs and a long list of specific rules that become obsolete, the IEC 61511 standard allows any design to be implemented. The standard allows the design to use old products or new technology. The standard allows innovation and good engineering practices. However, any design must be verified with documented performance metrics which must match risk reduction requirements in the form of SIL. In order to verify that a design meets the needed risk reduction, the designer must check three performance criteria [1]. *exida* calls these “the three barriers.”

The achieved SIL level of the SIF is the minimum of:

- Barrier 1 - SIL level based on Systematic Capability (SC) of each device used in a SIF. SC is a measure of design quality that shows sufficient protection against systematic design faults. SC is achieved either by choosing a certified part with SC to the given SIL level or greater or by completing a prior use justification to the given SIL level or greater. The lowest SC for any device in the SIF determines the SIL level for the SIF with respect to SC.
- Barrier 2 - SIL level based on minimum architecture constraints (SILac) for each element (subsystem) in a SIF. There are many different tables that can be used to establish architecture constraints; one in IEC 61511, and two alternatives are in IEC 61508 (Route 1H or Route 2H). The lowest SILac for any SIF subsystem determines the SIL level for the SIF with respect to SILac.
- Barrier 3 - SIL level based on a PFH (high demand), or a PFDavg (low demand) for the entire SIF.

All three of these design barriers must achieve or exceed the target SIL level. The worst case (lowest) SIL determines the SIL level for the SIF. Additionally, when RRF is specified the designer must ensure that $1/\text{PFD}_{\text{avg}}$ exceeds the required risk reduction factor (RRF).

Barrier Three: PFDavg Calculation

PFDavg calculation is an extremely important part of safety engineering in low demand applications as it is probably the **most difficult of the three barriers to meet** if realistic assumptions are made and if **realistic failure rates are used** (e.g., failure rates from www.SILSafeData.com). Target levels for PFDavg are defined in IEC 61508 for each of 4 levels of SIL. The highest safety is achieved in SIL 4 and the lowest in SIL 1. Table 1 shows that PFDavg for a given SIF will correspond to an equivalent SIL level within an order of magnitude range.

Safety Integrity Level	PFDavg Low Demand Mode of Operation
4	$\geq 10^{-5}$ to $< 10^{-4}$
3	$\geq 10^{-4}$ to $< 10^{-3}$
2	$\geq 10^{-3}$ to $< 10^{-2}$
1	$\geq 10^{-2}$ to $< 10^{-1}$

Table 1: SIL Level related to PFDavg

How can realistic values of PFDavg be calculated and what variables need to be taken into account when computing PFDavg?

PFDavg Key Variables

As a result of research into hundreds of sets of field failure data and proof test results, a number of things have been observed which may significantly impact a PFDavg. *exida* has compiled a list comprised of nine variables that must be considered in order to calculate a realistic and safe PFDavg. Failure rates of each device including failure modes and any diagnostic coverage from automatic diagnostics, λ_{DD} , λ_{DU} (attributes of the equipment chosen).

1. Mission Time, **MT** – the time period a set of equipment will be operated before overhaul or replacement (assignable by end user practices).
2. Proof Test Intervals, **TI** (assignable by end user practices).
3. Proof Test Effectiveness, **C_{pt}** (an attribute of proof test method).
4. Mean Time to Restore, **MTTR** (an attribute of end user practices) which includes **DTI**, Diagnostic Test Interval (an attribute of the device).
5. Proof Test Duration, **PTD** (an attribute of end user practices).
6. Probability of Initial Failure, **PIF** (an attribute of end user practices).

7. Site Safety Index, **SSI** (an attribute of end user practices).
8. Redundancy of devices including common cause failures (an attribute of SIF design).

Many of these variables are not commonly recognized and therefore not included in PFDavg calculations, yet these variables may impact the result by a SIL level or more.

Failure Rates, λ_{DD} , λ_{DU}

Failure rates, in particular the dangerous failure rates, come from a variety of sources [2, 3, 4]. Most manufacturers provide a third party FMEDA prediction that has been verified by fault injection testing and field failure analysis [5, 6].

When automatic diagnostics are designed into a device or subsystem, FMEDA analysis can distinguish between those failures detected and those undetected by the automatic diagnostics. The total dangerous failure rate, λ_D is partitioned into two subcategories: λ_{DD} , Dangerous Detected and λ_{DU} , Dangerous Undetected.

$$\lambda_D = \lambda_{DD} + \lambda_{DU}$$

This partitioning of λ_D into λ_{DD} and λ_{DU} is important because λ_{DD} and λ_{DU} contribute differently to PFDavg. In the event that there are no automatic diagnostics, λ_{DD} equals 0 and λ_{DU} equals λ_D .

Mission Time, MT

Mission Time is a period of time during which a set of equipment operates. This is an original reliability engineering term that is used to define the probability calculation period. Most end users choose a MT of 5, 10, 15, or 20 years which corresponds to the end of life for the process equipment or a period of time between each major shutdown and overhaul/replacement of all equipment. Any SIF device that reaches the end of its useful life during the MT is assumed to be replaced or completely overhauled and tested prior to or at the end of the device's useful life.

Assuming no automatic diagnostics and no proof testing during the MT, and given a dangerous failure rate and value for MT, an approximation for probability of failure on demand as a function of time, PFD(t), for a simplex (non-redundant) system can be shown to be:

$$\text{PFD}(t) = \lambda_{DU} * \text{MT}.$$

Under these conditions, the PFDavg is:

$$\text{PFDavg} = \lambda_{DU} * \text{MT}/2. \quad (1)$$

Proof Test Intervals, TI

First, the impact of ideal proof testing is explored. In most industrial applications where a Safety Instrumented System (SIS) is present, it is possible to design the SIF which comprise the SIS so that each SIF can be manually proof tested to see if it is working or not. The purpose of a proof test is to detect failures attributable to λ_{DU} because failure attributable to λ_{DD} will be identified by the automatic diagnostics if these are present. Again, assume no automatic diagnostics. If it is assumed that the proof test is 100% effective and requires no bypass time, this is called a perfect proof test. Now this assumption is quite unrealistic but is useful in showing the development of simplified equations to calculate PFD_{avg} . At the end of a perfect proof test, it is assumed that no SIF failures are present. This means that the probability of failure at the conclusion of the proof test is ideally zero. $PFD(t)$ with perfect proof testing looks like a repeating saw tooth as shown in Figure 1.

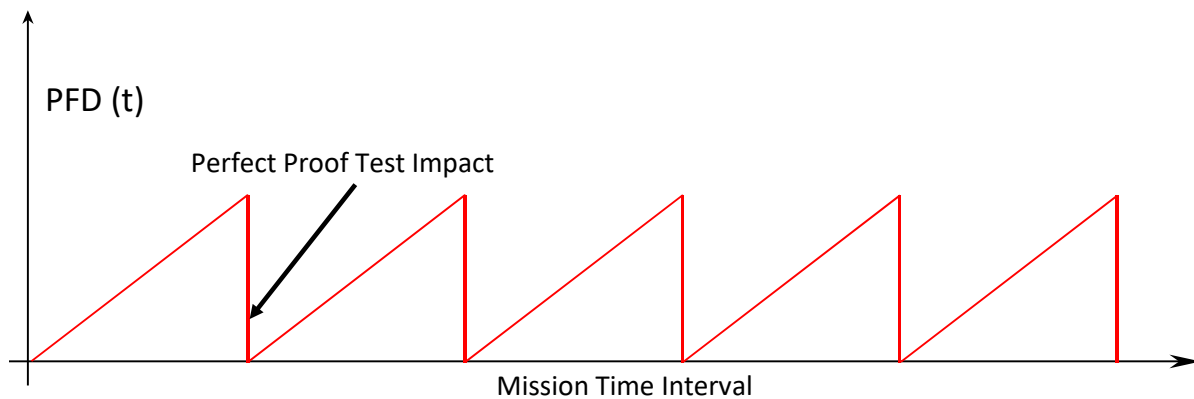


Figure 1: $PFD(t)$ showing multiple identical cycles with a perfect proof test.

The book *Safety Instrumented System Design – Techniques and Design Verification* [7], Chapter 4 explains the derivation of this plot in great detail and provides the equation for PFD_{avg} as:

$$PFD_{avg} = \lambda_{DU} * \frac{TI}{2}. \quad (2)$$

The MT is no longer a variable in this situation because the PFD_{avg} of each of the proof test cycles is the same as the PFD_{avg} of the first cycle as a result of perfect proof testing restoring the SIF to “as good as new” at the conclusion of each proof test. This equation for PFD_{avg} is, of course, very idealistic and unrealistic, but it is an appropriate place to start the development of more realistic models and equations.

Proof Test Effectiveness, C_{PT}

What happens in a real proof test? It can clearly be shown via detailed analysis of devices and examples that no real proof test is perfect [7, Chapter 12]. There are many examples of failures in products that cannot be detected by proof testing. An obvious example is a proof test performed by putting a blocking device on an actuator and checking to see if the actuator / valve assembly attempts to move. This does show that a portion of the subsystem is working but the test gives no indication of the health of many parts including the valve seat. Did the valve actually seal? This test cannot tell and is clearly not perfect.

What happens to PFD(t) when one performs an imperfect proof test? At the end of the proof test it is known that the probability of failure is reduced but it is not zero because not all failures are detected. Probability of failure is reduced to some value above zero. The probability of failure will increase after each proof test. This continues for the entire MT of the SIF. Figure 2 shows PFD(t) for an imperfect proof test.

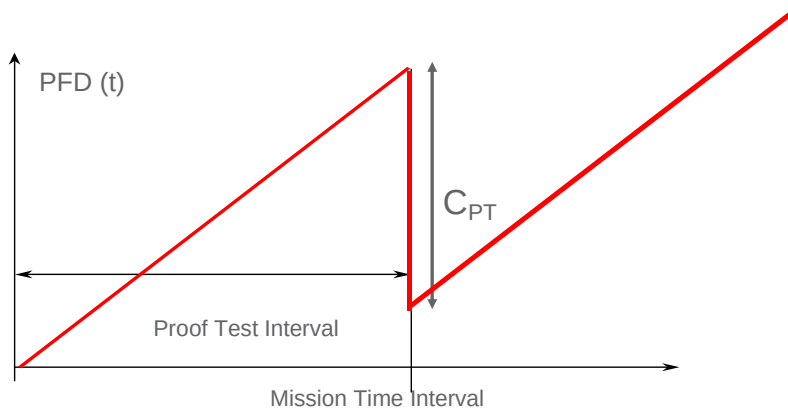


Figure 2: PFD(t) with imperfect proof testing.

Figure 3 shows the PFDavg for the entire MT consisting of six proof test intervals. Comparing the PFDavg of the first test interval with the overall PFDavg clearly shows a larger PFDavg for the entire MT. This difference is due to proof test effectiveness.

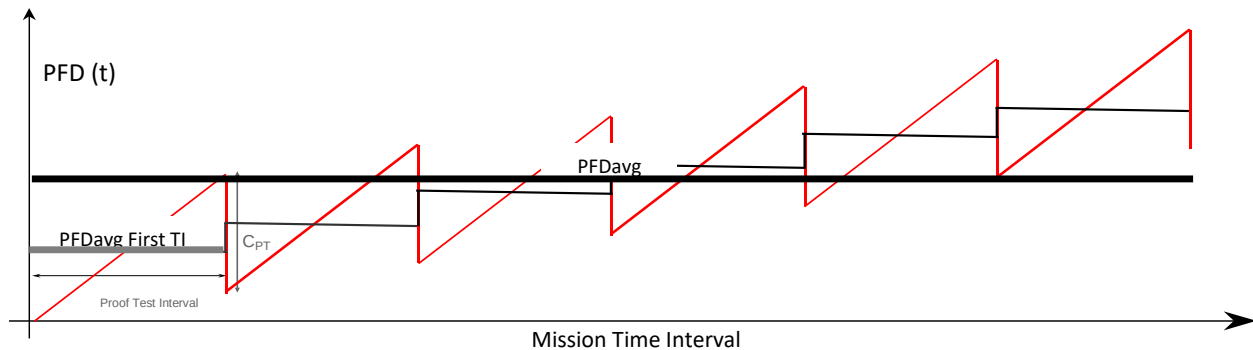


Figure 3: PFD(t) with imperfect proof testing showing PFDavg over the MT.

Proof test effectiveness can be expressed in a simplified approximate equation. The proof test effectiveness, C_{PT} , is a number between 0-100% which indicates the portion of the λ_{DU} detected by the manual proof test. The first term of the new equation approximating PFDavg uses the ideal formula for PFDavg multiplied by C_{PT} as those failures are detected by the proof test. The second term of the new equation shows failures not detected by the proof test $(1 - C_{PT})$ averaged over the MT.

$$PFDavg = \left[C_{PT} * \lambda_{DU} * \frac{TI}{2} \right] + \left[(1 - C_{PT}) * \lambda_{DU} * \frac{MT}{2} \right] \quad (3)$$

Mean Time to Restore, MTTR

MTTR, as applied to a SIF, accounts for all the time that the process is in operation and is unprotected by the SIF. This includes the time the SIF spends in dangerous failure before the failure is detected and identified, and all the time required to then return the SIF to its correct functionality so that it once again provides process protection. MTTR is usually divided into the mean time to detect the failure (MDT) and the mean time to repair the failure (MRT). MTTR may be different for detected and undetected failures and is usually designated with appropriate subscripts. Specifically,

$$MTTR_{DD} = MDT_{DD} + MRT_{DD}$$

$$MTTR_{DU} = MDT_{DU} + MRT_{DU}$$

For SIF failures detected by automatic diagnostics which do not result in an automatic shutdown of the process being protected by the SIF, the impacts of $MTTR_{DD}$ on PFD_{avg} must be accounted for. MDT_{DD} is defined in IEC 61508 as $DTI/2$. DTI is a parameter which indicates the worst-case time for the automatic diagnostics in a device to complete one full diagnostic scan. Occasionally a device has a significant DTI (e.g. 8 hours). When that happens, the DTI must be added to the MRT_{DD} . However, most devices have a DTI specification of less than one hour and some devices have high speed diagnostics which operate in less than one second. Therefore, DTI is usually negligible and neglected. Then the impact on PFD_{avg} of $MTTR_{DD}$ is

$$PFD_{avg} = \lambda_{DD} * MTTR_{DD}$$

$$PFD_{avg} = \lambda_{DD} * MRT_{DD} \quad (4)$$

For SIF failures detected during manual proof testing, MDT_{DU} is approximately $TI/2$ and this has already been accounted for in Equation 3. However, to derive Equation 3 it was assumed that undetected failures were found and repaired with the process shutdown. When a SIF is bypassed (process is operating) during a proof test, the PFD_{avg} is also impacted by MRT_{DU} and the impact is given by

$$PFD_{avg} = \lambda_{DU} * MRT_{DU} \quad (5)$$

When the impacts of Equations 4 and 5 are added to Equation 3 the result is

$$PFD_{avg} = \lambda_{DD} * MRT_{DD} + \left[C_{PT} * \lambda_{DU} * \left(\frac{TI}{2} + MRT_{DU} \right) \right] + \left[(1 - C_{PT}) * \lambda_{DU} * \frac{MT}{2} \right] \quad (6)$$

Proof Test Duration, PTD

PTD was not included in Equation (6) because the terms in Equation 6 apply to contributions to PFDavg caused of the presence of a dangerous failure. PTD does not impact PFDavg if the process is shutdown while the proof test is conducted. However, PTD must be accounted for when the SIF is on bypass during proof testing and the process continues to operate without the benefit of SIF protection. The effects of PTD when the SIF is on bypass are addressed separately in this section.

A SIF is likely to be put on bypass when the proof testing will (or might) cause a false trip of the process unit. What happens to PFD(t) during that bypass time? When a SIF is put on bypass it will not respond to a demand. The PFD(t) during the duration of the proof test period equals 1. This will cause the PFD(t) function to look like Figure 4, where PFD(t) is 1 for the duration of the proof test and then returns to the expected level.

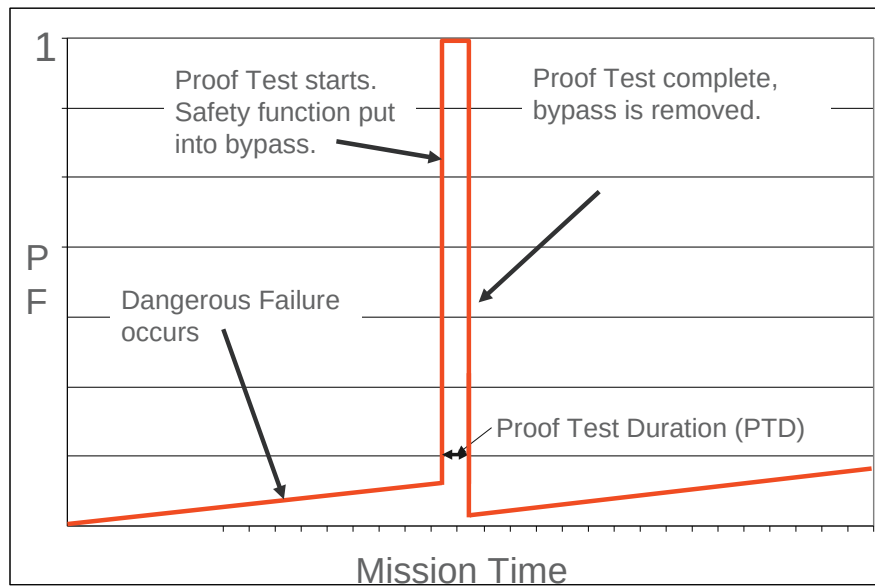


Figure 4: PFD(t) during a proof test bypass.

An additional term for the contribution to PFDavg of PTD can be easily developed if proof testing is performed during plant operation. The time the SIF spends on bypass during the proof test, PTD, occurs once every proof test interval, TI. If $MT = k \cdot TI$, then there will be $(k-1)$ proof tests during the SIF MT as a proof test will likely not be performed just before an overhaul. Therefore, the PFDavg attributable to the PTD will be

$$PFD_{avg} \approx (k-1) \cdot PTD / MT = ((k-1)/k) \cdot PTD / TI$$

However, approximating $(k-1)/k$ by 1 is conservative. Therefore, the contribution to PFDavg due to PTD is given more simply by:

$$PFD_{avg} \approx PTD / TI \quad (7)$$



Equation (7) can now be added to Equation 6 to create an equation that accounts for all variables so far considered:

$$PFDavg = \lambda_{DD} * MRT_{DD} + \left[C_{PT} * \lambda_{DU} * \left(\frac{TI}{2} + MRT_{DU} \right) \right] + \left[(1 - C_{PT}) * \lambda_{DU} * \frac{MT}{2} \right] + \frac{PTD}{TI} \quad (8)$$

Probability of Initial Failure, PIF

Due to the potential for human error or damage during commissioning or when complete commissioning testing cannot be performed after installation, there is a probability that an installed device will not work when the SIF initially is placed in operation. If this happens, PFD(t) is 1 at least until the first proof test. Now, if the initial failure is not discovered during commissioning testing, it may or may not be discovered during subsequent proof tests. In cases where the failure is not detected during subsequent proof testing, the initially failed SIF remains failed for the duration of its useful life!

The reader may feel that the probability of such an occurrence is negligibly small. However, that may not be correct. An extensive study of detailed proof test data [8, 9] where the proof test could detect the initial failure showed that there was clearly a PIF in some types of devices. Three independent data sets of pressure relief valves predicted an initial failure probability of approximately 1% – 1.6%! This accounted for the majority of failures observed in the population of proof test data.

The contribution of PIF to the approximate PFDavg calculation for a 1001 architecture requires only a small modification to Equation 8, and results in Equation 9, giving a conservative approximation for PFDavg, including six important variables, as follows:

$$PFDavg \approx PIF + (1 - PIF) [\lambda_{DD} * MRT_{DD} + C_{PT} * \lambda_{DU} * (TI/2 + MRT_{DU}) + (1 - C_{PT}) * \lambda_{DU} * MT/2 + PTD/TI] \quad (9)$$

Site Safety Index, SSI

In 1998, during a detailed study of field returns [10] at Moore Products, it was discovered that the return rate for identical modules differed by a factor of 4 (4X) from one site to another. Some failures were due to systematic problems where untrained people were damaging equipment during the proof test process. However, when those failures were removed from the data, there was still roughly a 2X difference in failure rate for the same device from site to site.

Since the 1998 study, several other field failure studies from a number of different sources, primarily end users in the process industries, have indicated there is also a difference in failure rates for the same product from site to site. Typically, the ratio is between 1.2X and 3X difference depending on product type.

Therefore, it can be concluded that random failures can be divided into two categories. There are random failures attributed to a product and random failures that are site specific. These seem to be related to procedures, training, and other variables that some have called the “safety culture.” *exida* defines this variable as the “Site Safety Index (SSI)” [11, 13, 14].

Several factors have been identified thus far which impact the SSI. These include the quality of:

1. Commissioning Test
2. Safety Validation Test
3. Proof Test Procedures
4. Proof Test Documentation
5. Failure Diagnostic and Repair Procedures
6. Device Useful Life Tracking and Replacement Process
7. SIS Modification Procedures
8. SIS Decommissioning Procedures
9. And others

SSI can be evaluated using a set of questions and a scoring system [12, 13]. The SSI model has five levels as shown in Table 2.

Table 2: Five levels of Site Safety Index from exSILentia

Level	Description
SSI 4	Perfect - Repairs are always correctly completed. Testing is always performed correctly and on schedule, equipment is always replaced before the end of useful life, equipment is always selected according to the specified environmental limits and process compatible materials, electrical power supplies are clean of transients and isolated, pneumatic supplies and hydraulic fluids are always kept clean, etc. This level is generally considered to be extremely difficult to achieve, but possible in some organizations.
SSI 3	Excellent -Repairs are correctly completed. Testing is performed correctly and on schedule, equipment is normally selected based on the specified environmental limits and a good analysis of the process chemistry and compatible materials. Electrical power supplies are normally clean of transients and isolated, pneumatic supplies and hydraulic fluids are mostly kept clean, etc. Equipment is replaced before the end of useful life, etc.
SSI 2	Good -Repairs are usually correctly completed. Testing is performed correctly and mostly on schedule. Most equipment is replaced before the end of useful life, etc.
SSI 1	Medium - Many repairs are correctly completed. Testing is performed and mostly on schedule, some equipment is replaced before end of useful life, etc.
SSI 0	Weak - Repairs are not always done. Testing is not performed; equipment is not replaced until failure, etc.

PIF, failure rates, probability of successful repair, probability of successful proof test, and probability of performing a proof test on schedule are all impacted by SSI because of the stochastic nature of those probabilities.

Redundancy/Common Cause

Redundancy has been used for many decades as a technique to improve both safety and availability of engineering systems. Appendix D in [7] shows various redundant architectures and PFDavg modeling techniques for those architectures. The detailed equations are beyond the scope of this paper. However, once redundancy is included, it is necessary to account for common cause failures. This is usually accomplished by use of β factors as described in IEC 61508 and IEC 61511. Equation 9 in this paper does not contain a common cause variable as it would not apply to a 1oo1 architecture.

Key Variables Summary

All of the variables listed need to be considered when calculating PFDavg for a SIF. These are contained in Table 3 along with indications of the source and applicability.

Table 3. Summary of Key Variables for PFDavg Calculations

Variable Number	Description	Source	Applicability
1	Failure rates, λ_{DD} and λ_{DU}	Manufacturer	Always
2	Mission Time, MT	End User	Always
3	Proof Test Intervals, IT	End User	Always
4	Proof Test Effectiveness, C_{PT}	End User	Always
5	Mean Time to Restore, MTTR	End User	For failures due to λ_{DD} , if automatic diagnostics do not trigger an automatic process shutdown For failures due to λ_{DU} , if proof testing is performed with process operating
5A	Diagnostic Test Interval	Manufacturer	Insignificant unless DTI is greater than one hour which does happen.
6	Proof Test Duration, PTD	End User	If proof test performed with process operating
7	Probability of Initial Failure, PIF	End User	If equipment not 100% tested after installation
8	Site Safety Index, SSI	End User	Always
9	Redundancy / Common Cause	System Designer	HFT ≥ 1

The Impact of Not Using Realistic Variables

To evaluate the impact on PFDavg of not using all important variables, consider the example of a high-level protection SIF. The proposed design has a SIL 2 target. The design uses a single SC 2 certified level transmitter, an SC 3 certified safety logic solver, and a single remote actuated valve. The actuated valve

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consists of a solenoid valve, a scotch yoke actuator and a ball valve all certified to SC 3. Using certified parts eliminates any need to perform prior use analysis for safety integrity purposes.

SIF Analysis with Optimistic Key Variable Values and Assumptions

The exSILentia tool accounts for all critical variables. For the first example using exSILentia, *idealistic/optimistic_variables* and assumptions are used to show the impact of neglecting all variables. The assumptions result in a model where only variables 1, 2, 3 and 5 listed in Table 3 are accounted for. MT is 5 years; the proof test interval is 1 year for the sensor and final elements, and 5 years for the logic solver; MTTR_{DD} is approximately MRT_{DD} which is assumed to be 8 hours. Variables 4, 6, 7, 8 and 9 were omitted from the calculations as follows. A proof test effectiveness of 100% is assumed which is the equivalent of omitting C_{PT} as a variable. It is also assumed that the proof test is performed with the process offline which sets MRT_{DU} and PTD to 0 in the calculations effectively omitting those variables. PIF is assumed to be 0 and SSI is set equal to 4. Finally, there is no redundancy so no common cause β factors apply.

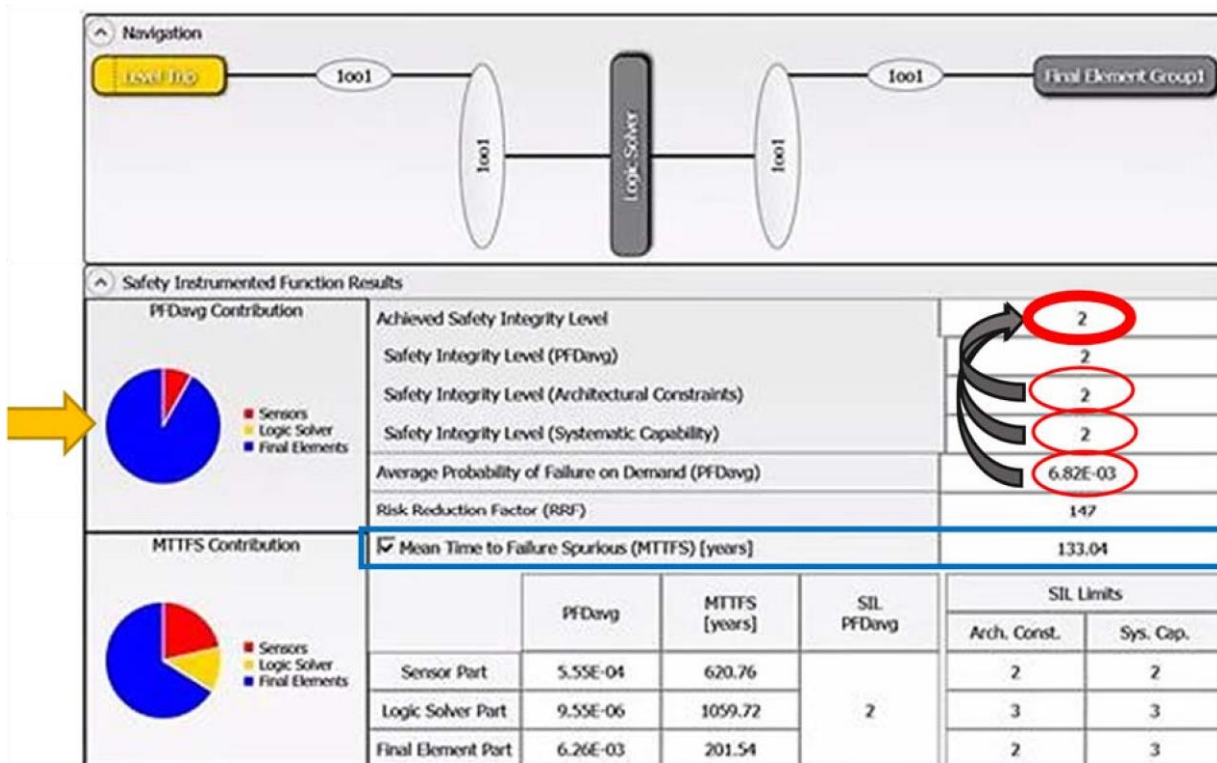


Figure 7: exSILentia Screen shot showing results of idealistic assumptions

Figure 7 illustrates the output from the analysis. From Figure 7, it is evident that the SIF systematic capability meets SIL 2. Further it can be seen that the architecture constraints meet SIL 2. Finally, in this example, the PFDavg was computed as 6.82×10^{-3} . This value meets SIL 2 with a Risk Reduction Factor (RRF) of 147. Therefore, the entire design meets SIL level 2 requirements for all three barriers (all indicated by red circles).

The pie chart on the left side of Figure 7 (indicated by an arrow) shows how much each SIF subsystem contributed to the PFDavg. The figure shows that final elements were the main contributor. The exSILentia tool also calculates the Mean Time to Fail Spuriously (MTTFS), which is boxed in blue. This number indicates how often a false trip will occur, so large numbers are the goal in order to avoid costly false trips.

SIF Analysis with Realistic Key Variable Values and Assumptions

But what if more realistic variables and assumptions were entered for the same SIF? In this more realistic scenario, a mission time of 25 years is used. A proof test interval of 1 year for the sensor and final element, as well as 5 years for the logic solver is used. Proof test coverage is reduced to 90% for the sensor and 70% for final element. MRT_{DD} is given a value of 48 hours which is more realistic. The proof testing is assumed to be performed with the process operating and the SIF on bypass. MRT_{DU} is equal to 48 hours and a PDT of 2 hours is included. PIF is still 0. A Site Safety Index of SSI 2 is used for the sensor and final elements, and SSI 3 for the logic solver. This second calculation considers *eight of the nine key variables*. Redundancy is not included as the architecture is 1001.

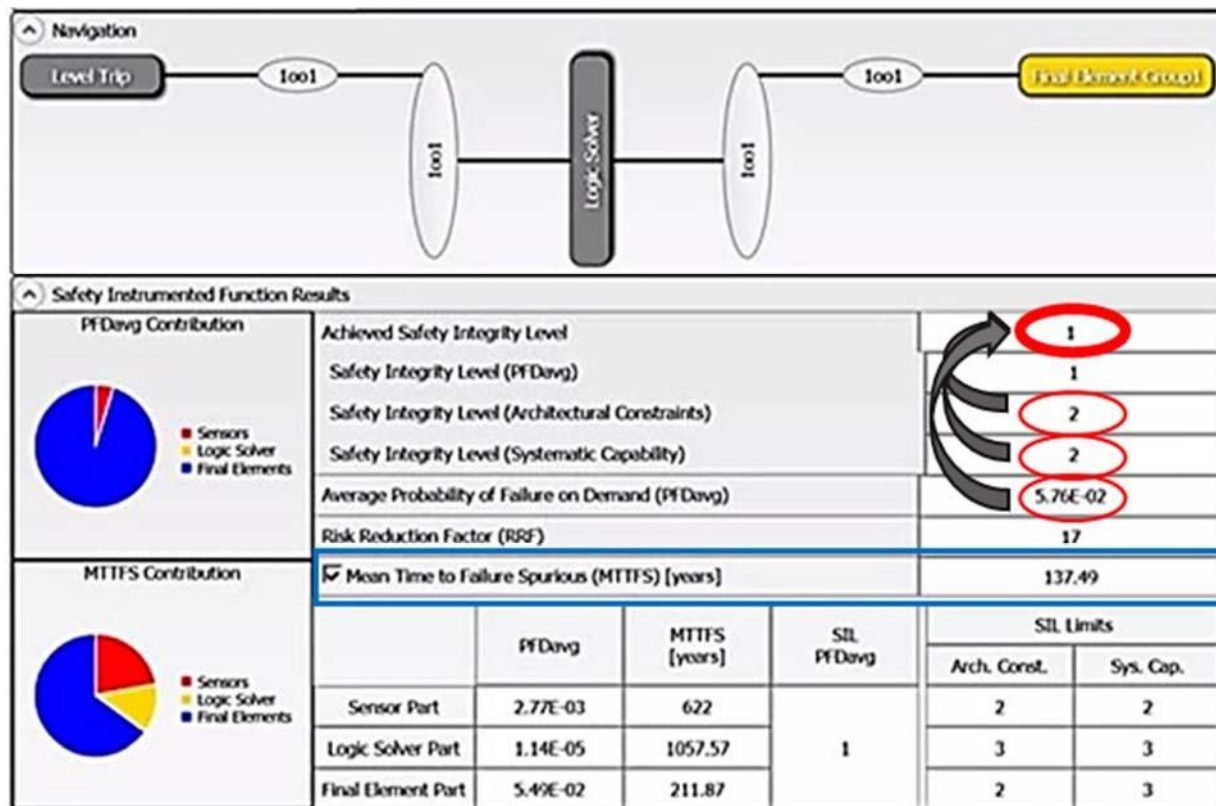


Figure 8: exSILentia screen shot with more realistic variables and assumptions considered

Figure 8 illustrates the output from the second analysis. The same design was re-analyzed but this time eight variables were included along with more realistic assumptions. What happened to the PFDavg? For the set of idealistic values and assumptions the PFDavg was 6.82×10^{-3} . The calculated PFDavg for this new analysis drops to a value of 5.76×10^{-2} ! The RRF, which was previously at a value of 147, now drops to 17! This barely meets the requirements for SIL level 1.

Why are these values so different? Sensitivity analysis indicates that C_{PT} is a significant variable. SSI is significant. The impact of PTD is not that significant in this case, but it sometimes can be.

Failure rates, redundancy, and proof test intervals are all well-known variables covered in IEC 61508, Part 6 equations. Proof test effectiveness is now a required variable for PFDavg calculations in IEC 61511. Other variables, including MT and especially Site Safety Index, are largely overlooked. All of the variables need to be taken into account to ensure a safe design.

Notation

CPT	proof test effectiveness also call proof test coverage
MDT	mean time to detect
MT	mission time
MTTFS	mean time to fail spurious
MTTR	mean time to restore
MRT	mean time to repair
nX	n times
PFDavg	average probability of failure on demand
PFD(t)	probability of failure on demand as a function of time
PIF	probability of initial failure
PTD	proof test duration
RRF	risk reduction factor
SC	systematic capability
SIF	safety instrumented function(s)
SIL	safety integrity level(s)
SILac	SIL based on architectural constraints
SIS	safety instrumented system(s); a SIS consists of one or more SIF
SSI	Site Safety Index
TI	proof test interval
1oo1	1-out-of-1 architecture; non-redundant architecture; single channel architecture
λ_D	constant failure rate for dangerous failures = $\lambda_{DD} + \lambda_{DU}$
λ_{DD}	constant failure rate for dangerous detected failures
λ_{DU}	constant failure rate for dangerous undetected failures

References

1. Three Steps in SIF Design Verification, White Paper, *exida*. Sellersville, PA www.exida.com, June 2014.
2. SINTEF, OREDA Offshore and Onshore Reliability Data Handbook, Vol 1. - Topside
3. Equipment and Vol. 2 - Subsea Equipment, 6th Ed, OREDA Participants, 2015.
4. Safety Equipment Reliability Handbook 4th Edition, *exida*. Sellersville, PA www.exida.com, 2015.
5. Bukowski, J. V. and Stewart, L. L., Explaining the Differences in Mechanical Failure Rates: *exida* FMEDA Predictions and OREDA Estimations, White Paper, *exida*. Sellersville, PA www.exida.com, July 2015.
6. Goble, W. M., and Brombacher, A. C., "Using a Failure Modes, Effects and Diagnostic Analysis (FMEDA) to Measure Diagnostic Coverage in Programmable Electronic
7. Systems," Reliability Engineering and System Safety, Vol. 66, No. 2, November 1999.

8. Grebe, J.C., and Goble, W. M., FMEDA – Accurate Product Failure Metrics, White Paper, *exida*. Sellersville, PA www.exida.com, V1.2, October 2009.
9. van Beurden, I. and Goble, W. M., Safety Instrumented System Design – Techniques and design Verification, ISA, Research Triangle Park, NC, 2018.
10. Bukowski, J. V., "Results of Statistical Analysis of Pressure Relief Valve Proof Test Data
11. Designed to Validate a Mechanical Parts Failure Database," Technical Report,
12. September, *exida*, Sellersville, PA, 2007.
13. Bukowski, J. V., and Goble, W. M., "Analysis of Pressure Relief Valve Proof Test Data," AIChE Journal Process Safety Progress, March 2009.
14. van Beurden, I.J.W.R.J., Reliability Analysis of Quadlog, Field failure research and study of the reliability information flow, Moore Products Co., Spring House, PA, USA, February 1998.
15. Bukowski, J. V. and Goble, W. M., "A Proposed Framework for Incorporating the Effects of End-User Practices in the Computation of PFDavg," *exida* white paper, January 2014. 12. Bukowski, J. V., Gross, R., and van Beurden, I., "Product Failure Rates vs Total Failure Rates at Specific Sites: Implications for Safety," Proceedings AIChE 11th Annual Global
16. Conference on Process Safety - Process Plant Safety Symposium, Austin, TX, April 2015.
17. Bukowski, J. V. and Chastain-Knight, D., Assessing Safety Culture via the Site Safety
18. IndexTM, Proceedings AIChE 12th Annual Global Congress on Process Safety -
19. ProcessPlant Safety Symposium, Houston, TX, April 2016.
20. Bukowski, J. V. and Stewart, L.L., Quantifying the Impacts of Human Factors on
21. Functional Safety, Proceedings AIChE 12th Annual Global Congress on Process Safety Process Plant Safety Symposium, Houston, TX, April 2016.

Revision History

Authors: Iwan van Beurden, William M. Goble, PhD

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Revision 1	First Release	October 1, 2015	
Revision 1.1	Updated SSI terminology	October 7, 2015	TES and WMG
Revision 1.2	Updated references, conditions	September 2016	WMG
Revision 2.0	Addition of Notation Section, New derivation of PFDavg, Changes to terminology to match terminology from IEC 61508, Updated references	April 2017	JVB
Revision 2.1	Added DTI, changed reference to new ISA book Feb 2018		WMG

exida – Who we are.

exida is one of the world's leading accredited certification and knowledge companies specializing in automation system cybersecurity, safety, and availability. Founded in 2000 by several of the world's top reliability and safety experts, exida is a global company with offices around the world. exida offers training, coaching, project-oriented consulting services, standalone and internet-based safety and cybersecurity engineering tools, detailed product assurance and certification analysis, and a collection of online safety, reliability, and cybersecurity resources. exida maintains a comprehensive failure rate and failure mode database on electrical and mechanical components, as well as automation equipment based on hundreds of field failure data sets representing over 350 billion unit operating hours.

exida Certification is an ANSI (American National Standards Institute) accredited independent certification organization that performs functional safety (IEC 61508 family of standards) and cybersecurity (IEC 62443 family of standards) certification assessments.

exida Engineering provides the users of automation systems with the knowledge to cost-effectively implement automation system cybersecurity, safety, and high availability solutions. The exida team will solve complex issues in the fields of functional safety, cybersecurity, and alarm management, like unique voting arrangement analysis, quantitative consequence analysis, or rare event likelihood analysis, and stands ready to assist when needed.

Training

exida believes that safety, high availability, and cybersecurity are achieved when more people understand the topics. Therefore, exida has developed a successful training suite of online, on-demand, and web-based instructor-led courses and on-site training provided either as part of a project or by standard courses. The course content and subjects range from introductory to advanced. The exida website lists the continuous range of courses offered around the world.

Knowledge Products

exida Innovation has made the process of designing, installing, and maintaining a safety and high availability automation system easier, as well as providing a practical methodology for managing cybersecurity across the entire lifecycle. Years of experience in the industry have allowed a crystallization of the combined knowledge that is converted into useful tools and documents, called knowledge products. Knowledge products include procedures for implementing cybersecurity, the Safety Lifecycle tasks, software tools, and templates for all phases of design.

Tools and Products for End User Support

- exSILentia® – Integrated Safety Lifecycle Tool



excellence in dependable automation

- PHAx™ (Process Hazard Analysis)
- LOPAx™ (Layer of Protection Analysis)
- SILAlarm™ (Alarm Management and Rationalization)
- SILect™ (SIL Selection and Layer of Protection Analysis)
- Process SRS (PHA based Safety Requirements Specification definition)
- SILver™ (SIL verification)
- Design SRS (Conceptual Design based Safety Requirements Specification definition)
- Cost (Lifecycle Cost Estimator and Cost Benefit Analysis)
- PTG (Proof Test Generator)
- SILstat™ (Life Event Recording and Monitoring)
- exSILentia® Cyber- Integrated Cybersecurity Lifecycle Tool
 - CyberPHAx™ (Cybersecurity Vulnerability and Risk Assessment)
 - CyberSL™ (Cyber Security Level Verification)

Tools and Products for Manufacturer Support

- FMEDAx (FMEDA tool including the exida EMCRH database)
- ARCHx (System Analysis tool; Hardware and Software Failure, Dependent Failure, and Cyber Threat Analysis)

For any questions and/or remarks regarding this White Paper or any of the services mentioned, please contact exida:

exida.com LLC

80 N. Main Street

Sellersville, PA, 18960

USA

+1 215 453 1720

+1 215 257 1657 FAX

info@exida.com